

Pulse Assessment

Evaluation Criteria — a plain-language guide to every check

How the assessment works

The assessment runs **111 checks** across **9 capabilities**. Each check measures a **coverage percentage** — how much of your estate has a given signal turned on:

$$\text{coverage \%} = (\text{entities WITH the feature}) \div (\text{total eligible}) \times 100$$

That percentage is compared to a **pass threshold**. Meeting it marks the check **passed**; otherwise it is **not met**. Two scores come out of this:

Coverage score	How many checks passed, as a simple average. Shown on the radar.
Maturity score	A weighted score by tier, with progressive gates (see below).

The three tiers — an adoption journey

- F** **Foundation** — the essentials everyone needs. Weight 60%.
- BP** **Best Practice** — deeper, correct usage. Weight 25% (counts only if Foundation ≥ 80%).
- E** **Excellence** — advanced / mature patterns. Weight 15% (counts only if Best Practice ≥ 60%).

The gates mean advanced wins don't inflate the score while the basics are still failing: you build a solid Foundation first, then depth, then sophistication.

About thresholds: most checks pass at ≥1% — the assessment rewards *any* adoption of a signal and lets the tier weighting tell the maturity story. A few use higher bars (diversity checks at ≥20-50%) where a minimum spread is the whole point.

1. Infrastructure Observability

Monitoring of hosts, processes, Kubernetes, and cloud platforms.

Tier	Check	What it validates	Pass
F	Host CPU coverage	Hosts reporting CPU metrics	≥1%
F	Host memory coverage	Hosts reporting memory metrics	≥1%
F	Host availability coverage	Hosts with availability metrics (SLA basis)	≥1%
BP	Host disk coverage	Hosts reporting disk utilisation	≥1%
BP	Host network coverage	Hosts with network interface entities	≥1%
BP	Process CPU coverage	Process groups with CPU metrics	≥1%
BP	Davis problem coverage	Hosts covered by Davis AI problem detection	≥1%
BP	Host-process topology	Hosts with detected process groups	≥1%
BP	Cloud host log enrichment	Hosts whose logs carry cloud-provider context	≥1%
BP	Cloud log enrichment	Log records enriched with cloud provider	≥1%
BP	Cloud account enrichment	Cloud logs carrying account ID	≥1%
BP	K8s node monitoring depth	Node-to-cluster ratio	≥1%
BP	Cloud namespace metric coverage	K8s namespaces with container metrics	≥1%
BP	Container restart monitoring	K8s namespaces tracking container restarts	≥1%
E	K8s cluster coverage	Clusters with active workload monitoring	≥1%
E	Host disk entity coverage	Hosts with disk entities detected	≥1%
E	Cloud workload coverage	K8s namespaces running cloud-native workloads	≥1%
E	K8s workload mapping	Clusters mapped to a cloud namespace	≥1%
E	Cloud region enrichment	Cloud logs carrying region metadata	≥1%
E	Cloud AZ enrichment	Cloud logs carrying availability-zone metadata	≥1%
E	Cloud span enrichment	Services with cloud context in spans	≥1%
E	Container resource limits	K8s namespaces with defined resource limits	≥1%

2. Application Observability

Distributed tracing and service RED metrics (Rate, Errors, Duration).

Tier	Check	What it validates	Pass
F	Service tracing coverage	Services with active span data	≥1%
F	Root span coverage	Services with incoming-request root spans	≥1%
F	Response time coverage	Services with p95 response-time metrics	≥1%
BP	Service method coverage	Services with method-level tracing	≥1%
BP	Failure tracking coverage	Services with failure-rate metrics	≥1%

Tier	Check	What it validates	Pass
BP	Throughput coverage	Services with request-count metrics	≥1%
BP	Service-process mapping	Services linked to process groups	≥1%
E	OTel instrumentation coverage	Services with OpenTelemetry spans	≥1%
E	Database span coverage	Services with database operation spans	≥1%
E	Messaging span coverage	Services with messaging spans (Kafka/RabbitMQ/SQS)	≥1%
E	Multi-service trace depth	Traces spanning 2+ services	≥1%
E	Service tagging maturity	Services with at least one tag	≥1%
E	Database call depth	DB services touching 2+ database systems	≥1%

3. Digital Experience

Real User Monitoring, Web Vitals, and synthetic monitoring.

Tier	Check	What it validates	Pass
F	RUM action coverage	Web apps with user-action metrics	≥1%
F	Session tracking coverage	Web apps with session tracking	≥1%
F	LCP coverage	Web apps with Largest Contentful Paint	≥1%
BP	INP coverage	Web apps with Interaction to Next Paint	≥1%
BP	CLS coverage	Web apps with Cumulative Layout Shift	≥1%
BP	Frontend error coverage	Web apps with error tracking	≥1%
BP	Synthetic HTTP coverage	Web apps covered by HTTP synthetic monitors	≥1%
BP	Synthetic availability coverage	Synthetic-monitor-to-web-app ratio	≥1%
E	Synthetic browser coverage	Web apps covered by browser synthetics	≥1%
E	Mobile app coverage	Apps with mobile monitoring	≥1%
E	Synthetic location diversity	Location-to-monitor ratio	≥1%

4. Log Analytics

Log ingestion, enrichment, and correlation quality.

Tier	Check	What it validates	Pass
F	Host log coverage	Hosts sending log data	≥1%
F	Service log coverage	Services with associated logs	≥1%
F	Log source diversity	Distinct log sources vs hosts	≥10%
F	Dedicated buckets usage	Grail buckets used for logs (≥2 = healthy)	≥50%
BP	Trace-correlated logs	Logs carrying trace_id	≥1%
BP	Entity-enriched logs	Logs with host entity context	≥1%
BP	Error log coverage	Services with ERROR-level logs	≥1%

Tier	Check	What it validates	Pass
BP	Log severity diversity	Severity levels ingested (of 5)	≥40%
BP	Structured logging	Logs with structured JSON	≥1%
BP	Multi-source host logging	Hosts logging from 2+ sources	≥1%
BP	Custom attribute enrichment	Logs with custom entity enrichment	≥1%
E	Process group log correlation	Logs with process-group context	≥1%
E	K8s log coverage	K8s namespaces with logs	≥1%
E	Log retention validation	Sources active 24h vs 2h	≥1%
E	Span-correlated logs	Logs carrying span_id	≥1%
E	Log-based events	Events generated from log data	≥50%

5. Application Security

Runtime vulnerability and attack-surface visibility.

Tier	Check	What it validates	Pass
F	Service security coverage	Services covered by security-event detection	≥1%
F	Runtime vulnerability baseline	Services with process groups (RVA prerequisite)	≥1%
F	Database interaction security	Services with DB tracing (SQLi surface)	≥1%
F	HTTP request surface coverage	Services with HTTP request tracing	≥1%
BP	Security event type coverage	Security event types vs expected categories	≥20%
BP	Error log security coverage	Services with ERROR-level logs	≥1%
BP	Warn log security coverage	Services with WARN-level logs	≥1%
BP	Attack detection coverage	Services with attack events detected	≥1%
E	Event kind diversity	Event kinds monitored (of 5)	≥40%
E	Failed request coverage	Services tracking failed requests	≥1%
E	Davis security problem coverage	Services covered by Davis AI	≥1%

6. Threat Observability

Davis problem detection and threat-log correlation.

Tier	Check	What it validates	Pass
F	Davis problem entity coverage	Hosts touched by Davis problems	≥1%
F	Security event service coverage	Services with security events	≥1%
F	Error log threat coverage	Hosts with ERROR-level logs	≥1%
BP	Problem entity correlation	Davis problems with affected entities	≥1%

Tier	Check	What it validates	Pass
BP	Problem category coverage	Problem categories detected (of 4)	≥25%
BP	Log source threat coverage	Distinct log sources vs hosts	≥10%
BP	Log entity enrichment	Logs with host entity context	≥1%
BP	Event entity correlation	Events carrying affected entity IDs	≥1%
E	Trace-correlated threat logs	Logs with trace_id for attack paths	≥1%
E	Recurring affected entity detection	Entities appearing in multiple problems	≥1%
E	Problem resolution coverage	Problems that reached CLOSED	≥1%

7. AI Observability

GenAI / LLM span monitoring (72h window for bursty workloads).

Tier	Check	What it validates	Pass
F	AI span service coverage	Services with AI/LLM spans	≥1%
F	Token tracking coverage	AI spans tracking token usage	≥1%
F	AI error tracking coverage	AI spans with error-status tracking	≥1%
BP	AI provider diversity	Known AI providers detected (of 5)	≥20%
BP	Agent invocation coverage	AI spans with agent-invocation tracing	≥1%
E	Prompt/response tracing coverage	AI spans capturing prompt or response	≥1%
E	Guardrail coverage	AI spans with guardrail monitoring	≥1%
E	Cost tracking coverage	AI spans with cost tracking	≥1%
E	AI tracing service breadth	AI-related share of total spans	≥1%

8. Business Observability

Business events (bizevents) linking telemetry to outcomes.

Tier	Check	What it validates	Pass
F	Service bizevent coverage	Services with business events	≥1%
F	Bizevent type diversity	Expected event types detected (of 10)	≥30%
F	Events with service context	Business events linked to services	≥1%
BP	Bizevent provider diversity	Expected providers detected (of 5)	≥20%
BP	Revenue data coverage	Events with revenue/monetary value	≥1%
E	Session-linked events	Events linked to RUM sessions	≥1%
E	Trace-linked events	Events correlated with traces	≥1%
E	Cost center coverage	Events with cost-center / product data	≥1%

9. Software Delivery

Deployment tracking and release-readiness signals.

Tier	Check	What it validates	Pass
F	Service deployment coverage	Services with deployment events (24h)	≥1%
F	Service request baseline	Services with active throughput	≥1%
F	Process group coverage	Services linked to process groups	≥1%
BP	Custom deployment coverage	Services with CI/CD deployment markers	≥1%
BP	Event kind diversity	Event kinds present (of 5)	≥40%
BP	Service failure baseline	Services tracking failure rate	≥1%
BP	Process group tagging	Process groups with ≥1 tag	≥1%
E	Event type diversity	Delivery event types detected (of 10)	≥30%
E	Davis problem detection	Services covered by Davis AI	≥1%
E	Ownership assignment	Services with ownership tags	≥1%

Totals at a glance

Capability	Checks	Foundation	Best Practice	Excellence
Infrastructure Observability	22	3	11	8
Application Observability	13	3	4	6
Digital Experience	11	3	5	3
Log Analytics	16	4	7	5
Application Security	11	4	4	3
Threat Observability	11	3	5	3
AI Observability	9	3	2	4
Business Observability	8	3	2	3
Software Delivery	10	3	4	3
Total	111	29	44	38

The app labels this "94 unique criteria" because several checks reuse the same denominator query (deduplicated at execution). The 111 above counts every scored check row across the 9 capabilities.

Companion documents: [DATA-SOURCES.md](#) (where each check's data comes from in Grail) and [SCORING-CALCULATIONS.md](#) (the full coverage-vs-maturity math). Version 2.5.3.